Project Work - Instructions

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1 Aim

The aim of the project work is to work independently, in a group of 2-3, on a project, which involves researching a topic, analysing and visualizing data, modeling with Python/Tensorflow and presenting results.

2 Grading

The project work contributes 30% to the final grade. Both the presentation and the code work are being evaluated.

3 Instructions

Please take note of the following instructions/guidelines:

- Explain the task to be solved.
- Point out, possibly, related work or problems in the literature/internet.
- Analyse your data. Visualize and explain the data features you deem to be relevant for the project.
- Explain your approach. That is, explain your model, the model architecture, optimizers, regularization, etc. used.
- Experiment with your model: change it, tune hyperparameters, etc. **Do not copy paste a model without substantially adopting it to your problem/task.** Explain your final model.
- Show/explain/visualize your results.
- Point out problems encountered.
- Point out further work or ideas.
- **Do not show your code in the presentation!** Rather explain everything in general terms, with figures, visualizations, etc.
- Due to time constraints, **the presentation is to take ca. 5 minutes**.
- The code needs to be clearly documented and handed in.

4 Schedule

Deadline to hand in the presentation and hand in the code and the presentation: 29.5.2026, 24:00.

5 Topic Examples

In principle you can work with the following data types:

1. Tabular data (FNN)
2. Image data (CNN)
3. Sequential data/time series data (RNN/FNN)
4. Text data (FNN/RNN)

Here, examples of topics for each of this category:

- Tabular data

  Detect credit card fraud
  `https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud`
  `https://keras.io/examples/structured_data/imbalanced_classification/`

- Image data

  Classify dolphins and whales
  `https://www.kaggle.com/competitions/happy-whale-and-dolphin/overview`

- Sequential/time series data

  Forecast air quality in Zurich
  `https://data.stadt-zuerich.ch/dataset/ugz_luftschadstoffmessung_stundenwerte`

- Text data

  Fake news detection
  `https://www.kaggle.com/c/fake-news/data`